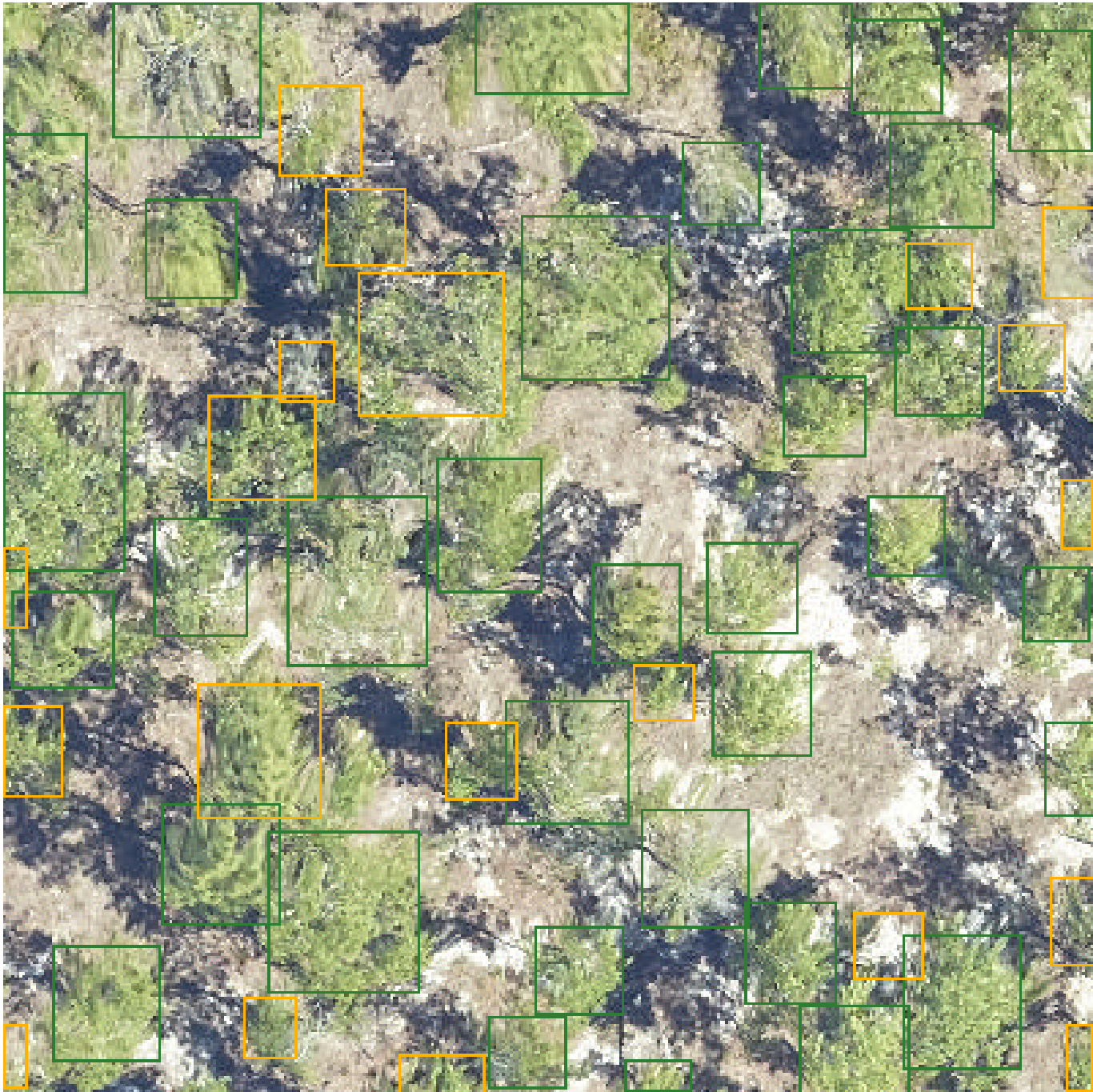


Forest Intelligence -- Analysis Report

Source image: OSBS_029.tif (NEON, Ordway-Swisher Biological Station) | Processed: 2026-09-13

Detection Snapshot



Forest Overview

Trees detected	55
Estimated canopy coverage	43%
Estimated canopy area	691 m^2
Reliability	MODERATE
Analyzed area	1,600 m^2

Summary

The analyzed area contains 55 detected tree crowns covering about 691 m^2, roughly 43% of the analyzed

area. Some regions are less certain - see the flagged review areas below.

Reliability Notes

- [!] 20 of 55 detections are below the confidence threshold (50%).
- [OK] Most crowns are clearly separable from their neighbors.

Review Areas

- 0 dense-overlap regions
- 20 low-confidence detections
- 0 shadow-affected regions

Known Limitations

- Crown detections are bounding boxes, not hand-segmented crown outlines - reported "canopy area" is the extent of each box, which tends to slightly overstate the true irregular shape of a crown.
- Physical-unit area figures (m^2) are only available when the source image carries usable spatial reference metadata; otherwise figures are pixel-based.
- Dense, overlapping canopy and heavily shadowed regions are flagged, but flagged $\neq$ fixed - those counts are still estimates.
- The pretrained model was trained primarily on specific forest sites and sensor types; performance on a very different forest type, altitude, or camera can vary and hasn't been independently validated here.
- This tool does not and cannot determine species, age, biomass, ecological health, biodiversity, or carbon stock/sequestration from crown imagery alone - those require additional data and validated methodology this tool does not have.
- The shadow-affected flag is a brightness heuristic, not true shadow detection - it's a useful signal, not a certainty.

This report does not estimate species, biomass, ecological health, or carbon stock/sequestration -- these require data and methodology beyond crown detection and are not claimed here.